BSCY PASSWORD-BASED DIGITAL ACCESS CONTROL SYSTEM

A Project Report

on

Password-Based Digital Access Control System

By

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A Project Report submitted to the

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in partial fulfillment of the requirements for the course of

DIGITAL LOGIC DESIGN (DLD)

Faculty of Engineering

Capital University of Science & Technology,

Islamabad

June, 2026

DECLARATION

It is declared that this is an original piece of our own work, except where otherwise acknowledged in text and references. This work has not been submitted in any form for another degree or diploma at any university or other institution for tertiary education and shall not be submitted by us in future for obtaining any degree from this or any other University or Institution.

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CERTIFICATE OF APPROVAL

It is certified that the project titled "**Password-Based Digital Access Control System**" carried out by Aetisaam ul Hassan, Reg. No. BCY253051, Aman Sajid, Reg. No. BCY253084, and Maier Masood, Reg. No. BCY253033, under the supervision of the course instructor, Capital University of Science & Technology, Islamabad, is fully adequate, in scope and in quality, as a term project for the course Digital Logic Design (DLD).

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We would like to thank our respected course instructor whose guidance helped to complete our report and practical implementation in the given time. Last but not the least, this project cannot be completed without the effort and co-operation of group members.

ABSTRACT

This report discusses about the design and testing of a Password-Based Digital Access Control System by using Digital Logic Design concepts. The aim of this project is to allow access only when the correct 4-bit password is entered through input switches on a breadboard. The system is designed in two separate modes: normal user mode and admin mode. Each mode has its own stored password, and the entered password is compared with the selected stored password by using XOR gates followed by NOT gates to form equality logic. The equality outputs of all four bits are combined through cascaded 2-input AND gates, because the project does not use any 4-input gate. The normal user match and admin match are then selected through mode selection logic and combined to generate the final match signal. A 74LS74 D flip-flop is used to latch the access granted output after the ENTER button is pressed, while another flip-flop section is used for the wrong password alarm. If the selected password matches, the green LED turns ON and shows access granted. If the password is wrong, the red LED turns ON and the buzzer rings to indicate access denied. The whole circuit is implemented with TTL ICs such as 74LS86, 74LS04, 74LS08, 74LS32 and 74LS74, along with switches, resistors, LEDs, buzzer, jumper wires and regulated 5V supply. This device shows the practical use of truth tables, Boolean expressions, combinational logic, sequential logic, IC wiring, and breadboard-based circuit implementation. The final project is suitable for demonstrating digital security logic without using any microcontroller, motor, relay or software-controlled unlocking mechanism.

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LIST OF ACRONYMS/ABBREVIATIONS

DLD	Digital Logic Design
IC	Integrated Circuit
TTL	Transistor-Transistor Logic
DIP	Dual In-line Package
LED	Light Emitting Diode
DC	Direct Current
VCC	Positive Supply Voltage
GND	Ground
XOR	Exclusive OR
XNOR	Exclusive NOR
FF	Flip-Flop
PCB	Printed Circuit Board

Chapter 1

INTRODUCTION

The password-based digital access control system is basically a digital security project used to verify a binary password with the help of logic gates and integrated circuits. This project is designed without using a microcontroller and without using a motor or relay. The complete decision-making process is performed through combinational logic and the final output is stored through sequential logic. The system allows access when the entered password is equal to the selected preset password and gives an alarm when the entered password is incorrect.

1.1 Overview

Our project is to get an appropriate digital output from a 4-bit password entered through switches. The working of the password comparison circuit and its result is our main aim. The entered password is compared with normal user password and admin password. The result is further shown by a green LED for access granted and by a red LED with buzzer for access denied. The output is latched with D flip-flop so that the result remains stable after the ENTER button is pressed.

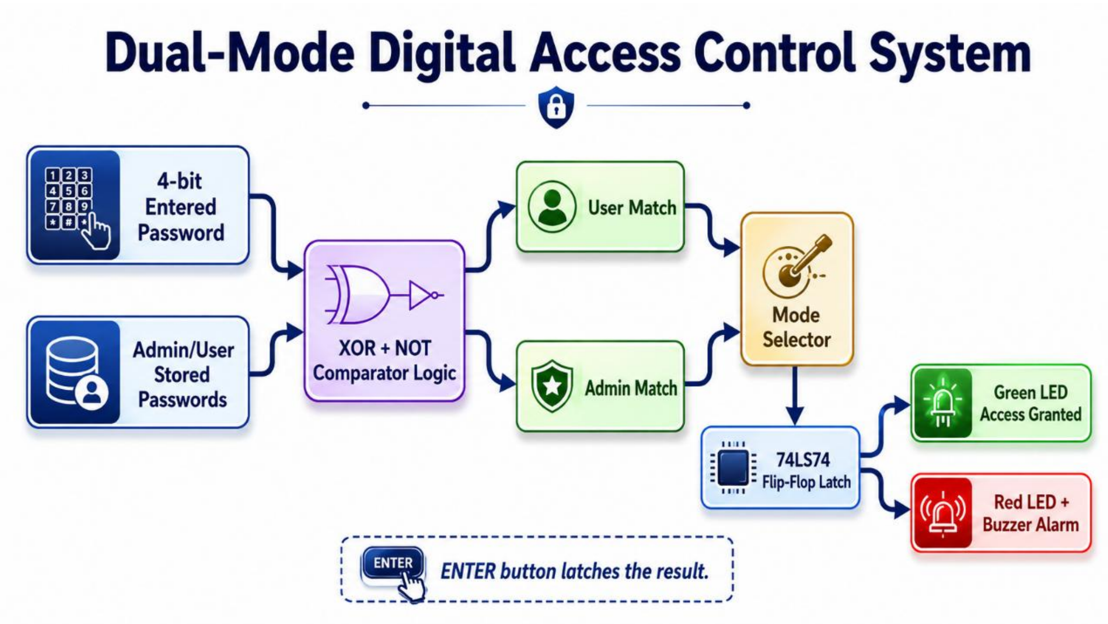

Figure 1 Model of digital access control system

1.2 Project Idea

The project idea is to design a digital password verification system which works on a breadboard. This system checks the password entered by the user and compares it with the stored password. There are two different modes in the project. In normal mode, the circuit checks the normal user password. In admin mode, the circuit checks the admin password. The correct password turns ON the access indicator and the wrong password activates the alarm path.

1.3 Purpose of the Project

The purpose of this project is to practically implement the concepts of Digital Logic Design. The project demonstrates binary inputs, truth tables, Boolean expressions, XOR/XNOR comparison, cascaded AND logic, mode selection, D flip-flop latching, LED indication and buzzer alarm. By working on this project, the complete logic from input to output is verified on a breadboard.

1.4 Project Specifications

The specifications of our project are that we are using a digital logic gate-based technique which is suitable for a DLD lab project. As there are many password locks based on microcontrollers, this project is prominent because it uses only TTL ICs and basic logic concepts. The circuit detects correct and wrong password conditions according to the selected mode.

Table 1 Project Specification

Sr.no	Objective	Specification
1	Logic Design	Combinational and sequential digital logic
2	Password Length	4-bit binary input password
3	Operating Modes	Normal user mode and admin mode
4	Comparator	XOR followed by NOT gates for equality checking
5	Match Logic	Cascaded 2-input AND gates using 74LS08
6	Output Latching	74LS74 D flip-flop for access and alarm
7	Indicators	Green LED, red LED, and buzzer
8	Power	Regulated 5V DC supply for TTL ICs

1.5 Applications of the Project

The following are the application of this project:

1.5.1 Secure access control

Basically, we are going to make a device that is going to check the password with the help of switches. Secondly, we have to check whether the input password is correct or wrong by using logic gates. The complete circuit can be used as a digital door lock demonstration, locker security circuit, restricted lab entry indication system, alarm system and DLD learning project. The project shows how a real access control decision can be made using only binary logic.

1.6 Project Plan

Give details of project plan. Project plan will contain details of how work is divided into part I & II and how resource persons have been allocated. Use tables to give details of project plan.

1.6.1 Project Milestones

Following is the distribution of tasks, task duration and resource person details for part I and II separately

Table 2 Project Milestone (Part1)

S#	Tasks	Duration	Resource Person
1	Literature Review	02 Weeks	Aetisaam
2	Truth table and logic design	02 Weeks	Aetisaam,Aman,Maier
3	Boolean expression simplification	02 Weeks	Aetisaam,Aman,Maier
4	Proteus simulation of password logic	02 Weeks	Aetisaam
5	IC and component selection	01 Week	Aetisaam,Aman,Maier
6	Reporting	1 week	Aetisaam,Aman,Maier

Table 3 Project Milestone (Part 2)

S#	Tasks	Duration	Resource Person
1	Breadboard layout planning	02 Weeks	Aman
2	DIP switch and resistor wiring	02 Weeks	Aetisaam,Maier
3	Comparator IC implementation	03 Weeks	Aetisaam,Aman,Maier
4	Flip-flop, ENTER and RESET wiring	01 Week	Aetisaam,Maier
5	Buzzer and LED output testing	02 Weeks	Aetisaam,Aman,Maier
6	Report writing and final testing	1 week	Aetisaam,Maier,Aman

1.6.1 Project Timeline

The timeline of our project is following:

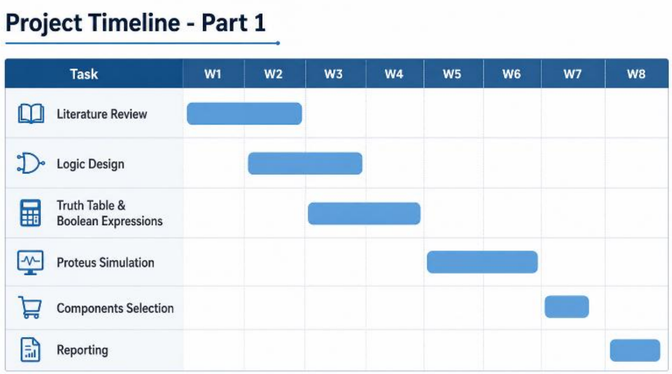

Figure 2 Part 1 timeline

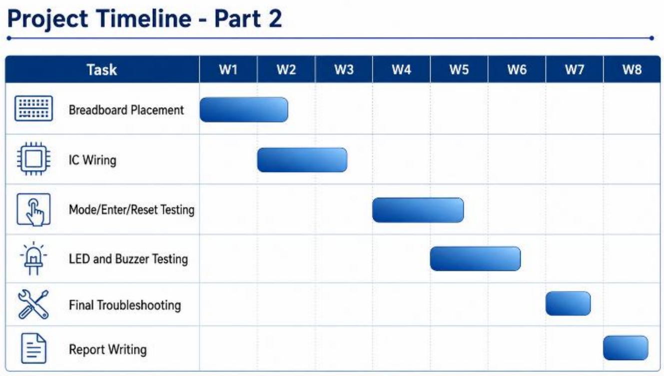

Figure 3 Part 2 timeline

1.7 Report Organization

There are specific chapters inside the record. First chapter is the introduction that offers the overview, motive and specifications of the project. Literature review is another chapter, which is about background theory, related technologies and related projects. Design and implementation give information about the design of project in hardware and logic simulation and it is applied and complete method of the project. Tools and techniques are about the one of kind equipment that are beneficial in the completion of project and what techniques are used to achieve the tasks. The technique using in this project is digital logic comparison through XOR, NOT, AND, OR and flip-flop ICs. This report is organized in such a way that it briefly shows about the simulation and hardware working. Results is the chapter from wherein consequences of the project can be checked that what is said at the begin did it completed or not. Conclusion is that what has been concluded from the final working prototype.

Chapter 2

LITERATURE REVIEW

2.1 Background Theory

In the background, there are many other access control systems using different techniques. In those techniques, microcontrollers, keypads, RFID modules and relays are commonly used to allow entry. These technologies are not purely based on Digital Logic Design. In this project, we are using a direct gate-level password verification technique. The complete decision is generated through logic gates and the final result is held using a flip-flop.

2.2 Digital Access Control

Digital access control is the method of allowing or denying access based on a binary decision. In a DLD circuit, the binary values are represented by logic 0 and logic 1. A password-based access circuit receives a set of input bits and compares them with a stored pattern. There are two basic access results.

- Access granted condition
- Access denied condition

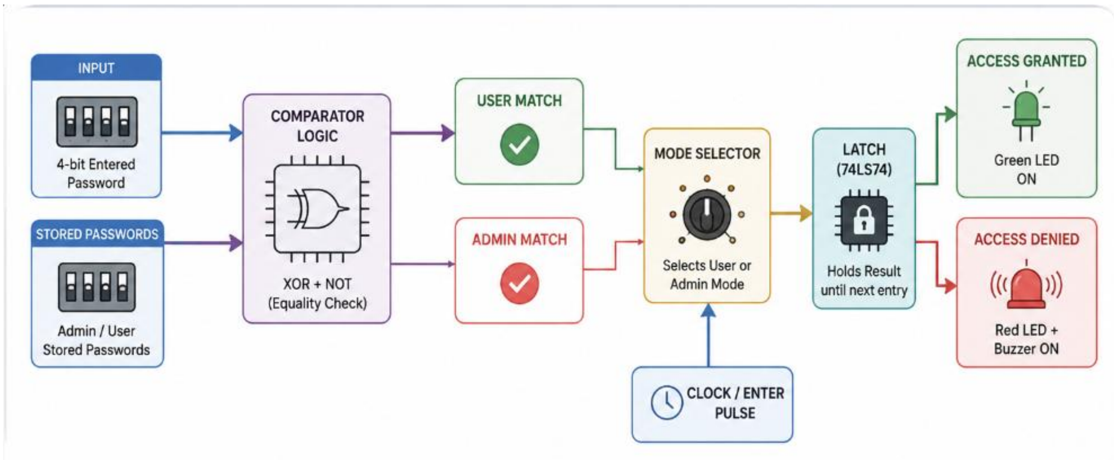

Figure 4 System concept diagram

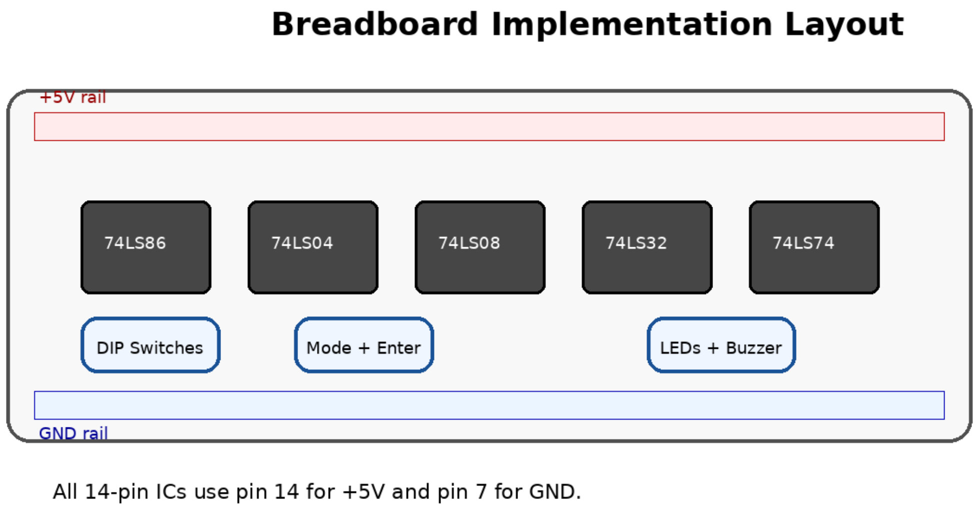

Figure 5 Breadboard implementation layout

2.3 Password Verification Logic

Password verification is an approach to find whether the input password is exactly equal to the stored password. The password detection identifies or recognizes the presence of correct combination whereas result classification classifies the input as correct or wrong. Digital logic is useful to perform this type of classification because each bit has only two possible values. The different logic gates applied for password verification are XOR, NOT, AND and OR. The XOR gate finds mismatch between two bits and NOT gate converts it into equality output. The equality outputs are combined to produce a final match.

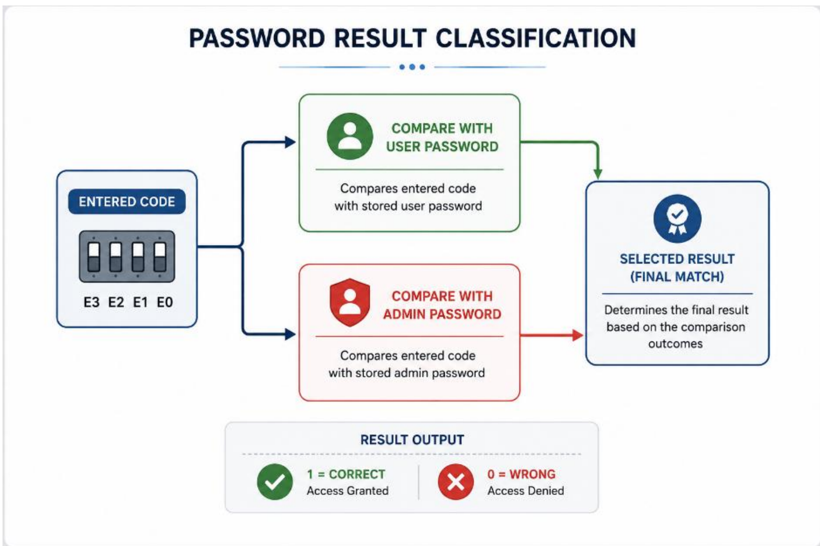

Figure 6 Password result classification

2.3.1 Classification

There are the many types of logic stages. It may be combinational stage as well as sequential stage or indication stage.

2.3.1.1 Combinational stage:

Combinational stage is the part of the circuit in which output depends only on present input. In this project, XOR, NOT, AND and OR gates are used as combinational logic. This stage compares password bits, combines equality signals and selects the final mode path. If any entered bit does not match with the stored bit, the match output becomes zero.

2.3.1.2 Sequential stage:

Sequential stage stores the output using a memory element. In this project, 74LS74 D flip-flop is used to latch the access granted and alarm results. The ENTER push button is applied as a clock input so that the output changes only after the user confirms the password. RESET is used to clear the output before the next attempt.

2.3.1.3 Indication stage:

Indication stage is the final stage of the project. A green LED shows correct password. A red LED with buzzer shows wrong password. A yellow LED may be used for admin mode indication. Current-limiting resistors are connected with LEDs to protect them from excess current.

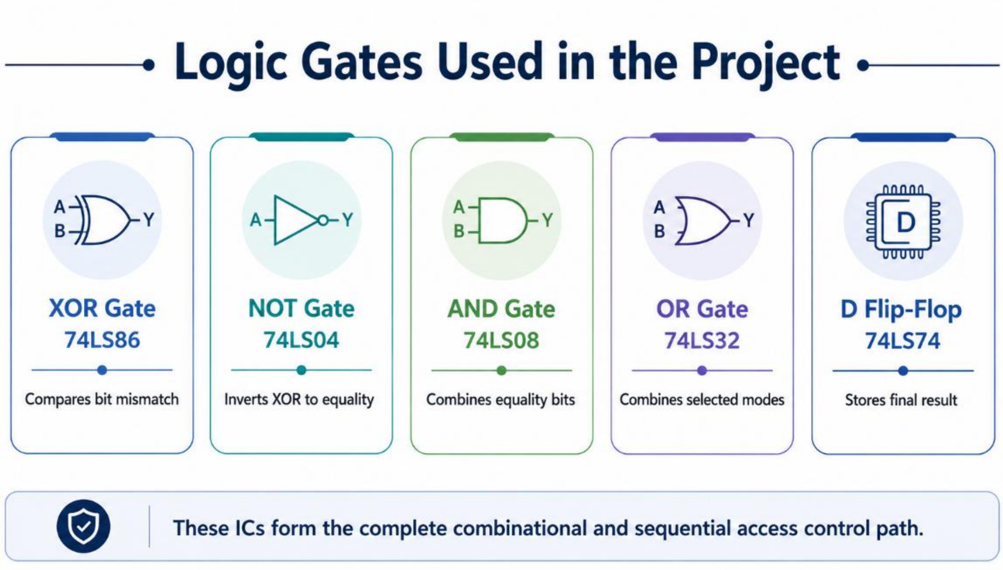

Figure 7 Logic gates used in the project

2.4 Related Technologies

The related technologies in order for access control were many possible techniques with different complexity. Then gate-level digital design is used for the purpose of understanding basic security logic. These technologies are truly based on binary layers which are interconnected with each other. This digital architecture includes combinational logic and sequential logic. These architectures are described below:

2.4.1 Combinational Logic Gates:

Combinational logic gates are the basic building blocks of the circuit. In this project, XOR gates compare two bits. NOT gates are used to invert the XOR outputs and generate XNOR/equality signals. AND gates combine equality signals to form a complete 4-bit match. OR gate combines the selected user and admin paths to obtain the final match output.

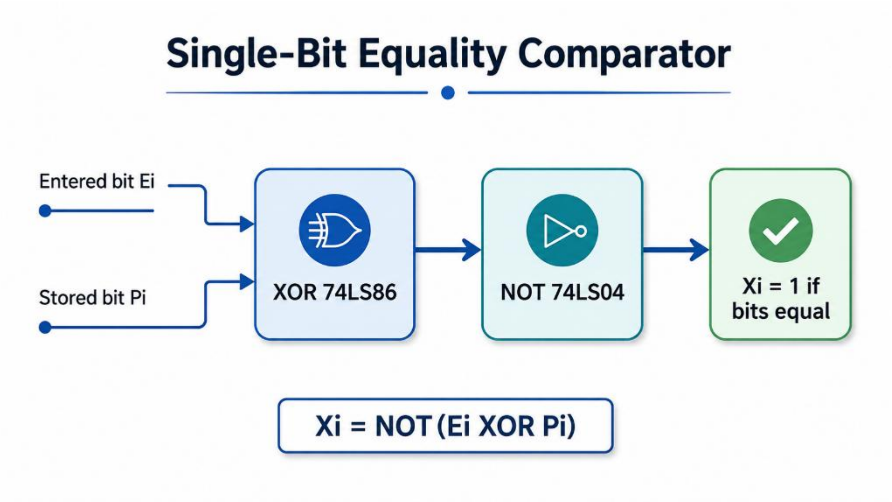

Figure 8 Single-bit equality comparator

2.4.2 Sequential Logic and Flip-Flops

A D flip-flop is a sequential device that stores one bit of data. In the project, the D input receives the final match or wrong condition and the ENTER button provides the clock. This makes the output stable and avoids flickering.

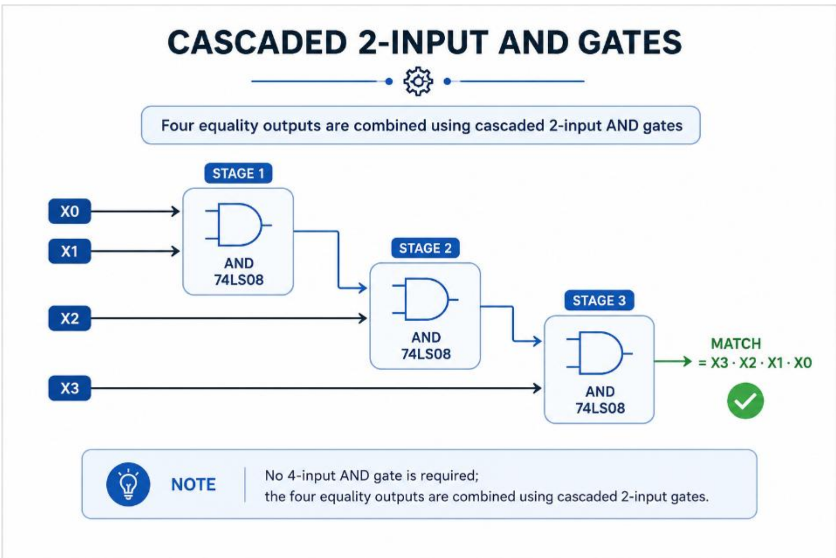

Figure 9 Cascaded AND gate matching logic

2.4.3 DIP Switches

A DIP switch is used as a manual binary input device. Each switch can provide logic 1 or logic 0 to an IC input. In this project, DIP switches are used for entered password bits, normal user password bits and admin password bits. Pull-down resistors are used to keep the input at logic 0 when the switch is OFF.

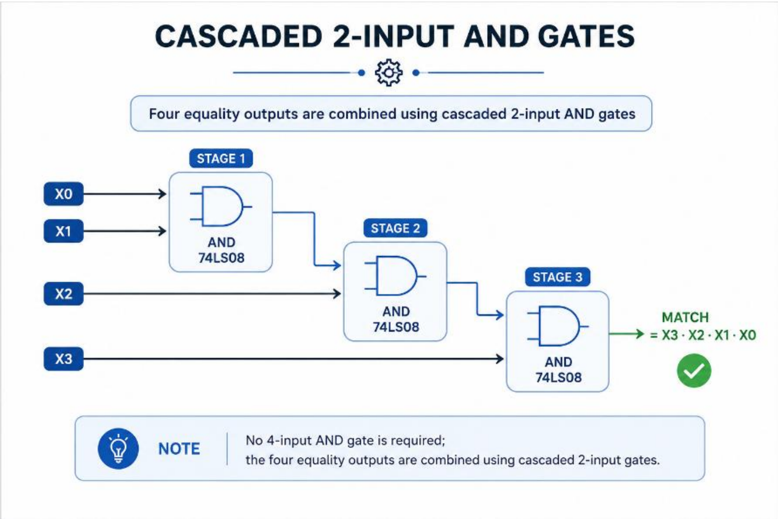

Figure 10 Mode selection logic

2.4.4 TTL Logic ICs

TTL logic ICs are used to build the circuit on breadboard. The 74LS series ICs work on 5V supply. The power convention for each 14-pin TTL IC is pin 14 connected to +5V and pin 7 connected to ground. Decoupling capacitors are recommended near IC power pins for stable operation.

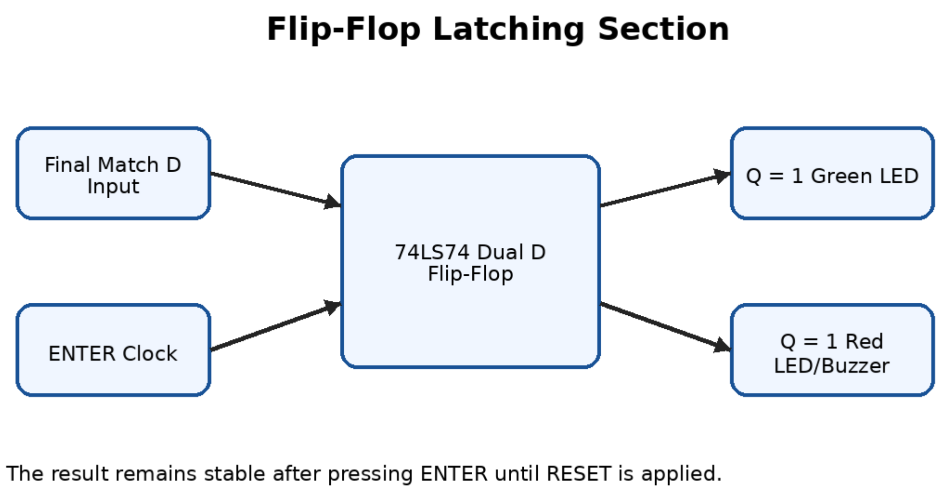

Figure 11 Flip-flop latching section

2.4.5 Simulation and Breadboard Testing

The major advantage of the logic simulation and breadboard analysis is that by using the simulation technique it provides verification before practical wiring. The processing difficulty of the project completely depends on the number of gates, the number of input lines and the clarity of wiring. After simulation, the same logic is connected on a breadboard and tested with actual switches, resistors, LEDs and buzzer.

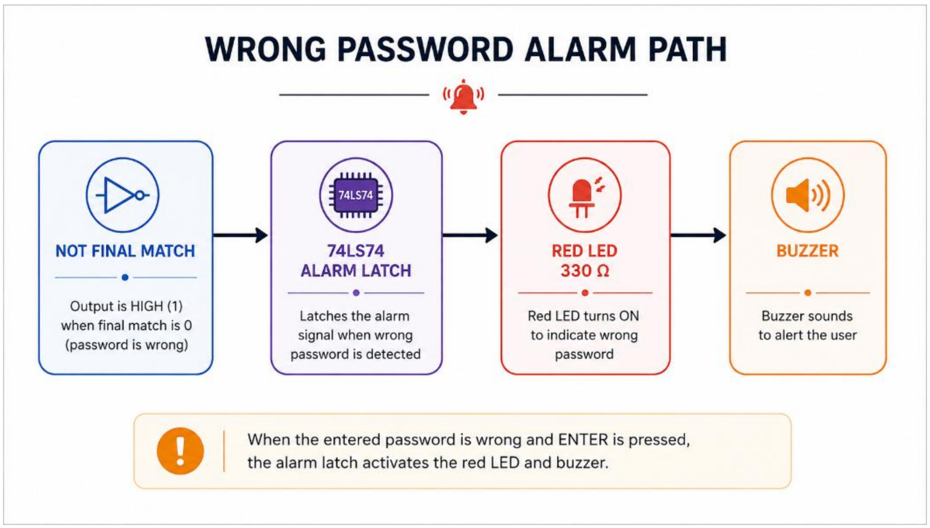

Figure 12 Wrong password alarm path

2.5 Related Projects

Following are the related projects for this project:

2.5.1 Password Based Door Lock using Logic Gates:

The related projects available for access control are generally based on password-based door locks. Some systems use microcontrollers and keypads. However, the DLD version can be made by using gates only. The input switches act like password keys and the logic ICs compare them with preset values. This approach is simple and cost effective for learning Boolean logic and IC wiring.

2.5.2 Digital Access Control System:

A digital access control system may include a password unit, comparison circuit, mode selector and alarm unit. In our project, the input password is compared with both user and admin password lines. The mode input decides which match is accepted. If the selected match is true, access is granted. If the selected match is false, the alarm circuit is activated. This design is suitable for DLD practical implementation because it does not require programming.

2.6 Related Studies/Research

To reduce the ratio of unauthorized access in basic security systems, password verification is one of the common ideas. In digital logic learning, the same concept is useful because it converts a real security problem into truth tables and Boolean expressions. This project shows how theoretical DLD concepts can be converted into working hardware.

2.7 Limitations and Bottlenecks of the Existing Work

The limitations for the existing work are the size of the breadboard circuit, number of wires, switch bouncing and power stability. If inputs are floating, the ICs can give random output. To overcome this, pull-down resistors are used with switches and push buttons. The circuit also needs careful placement of IC VCC and GND pins. For betterment in result, the password length can be increased or a keypad encoder can be added in future.

2.8 Problem Statement

Our problem statement is to make a digital access control system for password verification. A simple lock may allow unauthorized access if there is no password checking. In order to detect correct and wrong password conditions, we are using digital logic gates. The entered password is compared with the selected stored password. Therefore, we can get familiar with access control logic and reduce unauthorized entry in demonstration form. The circuit is implemented on breadboard, and the output is displayed using LEDs and buzzer.

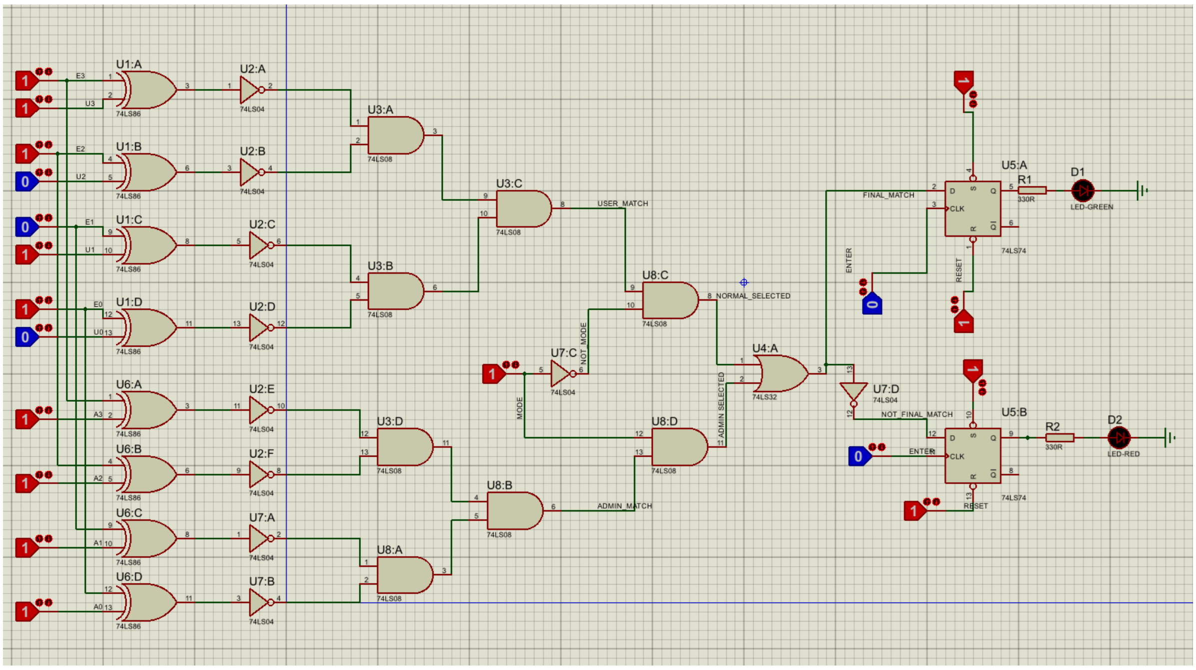

Figure 13 Final circuit diagram

2.9 Summary

This chapter covers all the literature survey that includes all the work that is required for this project. It also covers the background theory of password verification and IC-based digital logic. This chapter also includes the related technology to our project that shows the same working field as our project is implementing. Furthermore, related projects were also added in a chapter that shows the output of some particular working prototypes. The limitations and bottlenecks of the related products are also discussed which are tried to be compensated in this project. The problem statement of the project is also discussed in detail in this chapter.

Chapter 3

PROJECT DESIGN AND IMPLEMENTATION

The project mainly consists of two designs; the hardware design and the other design is based on logic simulation. Both these designs are discussed in this chapter with their implementation procedures.

3.1 Proposed Design Methodology

This project used a prototype for password-based digital access control. This prototype is small in size as compared to a real door lock security system. Basically, the switches are used to enter the live binary password and the circuit compares it with preset password values. The access decision is controlled by the mode selection switch. The movement of data from input to output is controlled by gates and flip-flops only.

3.1.1 Model Designing:

The project main backbone is its digital logic model. The model is based on comparator logic in order for password detection system. This detection system logic is specifically based on binary password comparison. Our architecture is based on following:

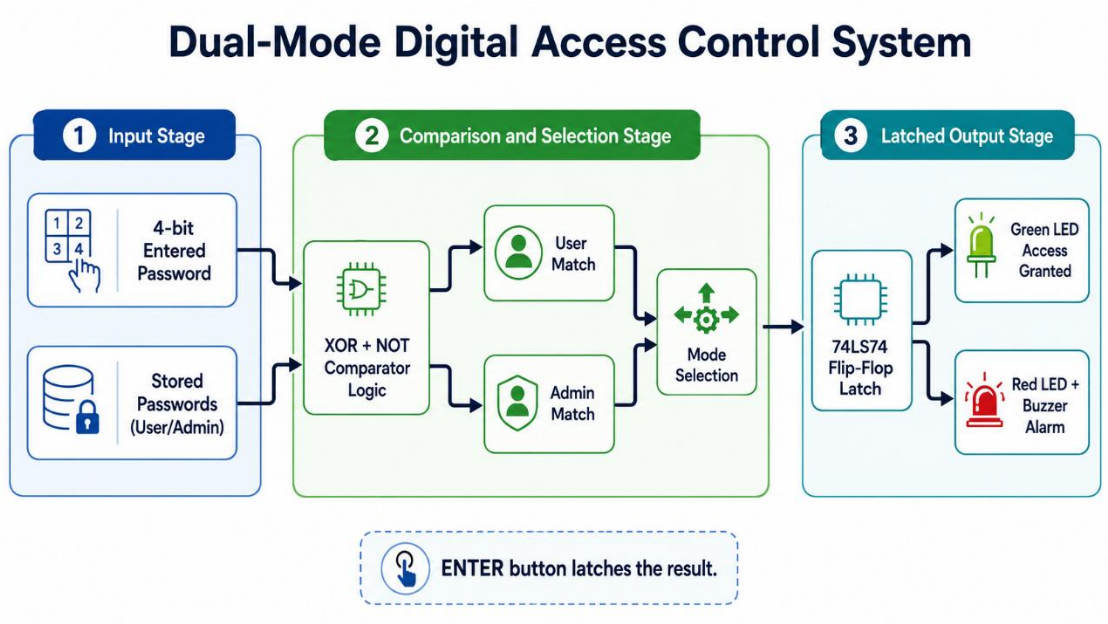

Figure 14 Block Diagram

3.1.1.1 Input Binary Password:

The binary password is the main part of our project, through which the circuit can differentiate either the access is correct or wrong. The input password is directly taken from the DIP switch which is connected with logic IC inputs. The input is further processed through comparator logic.

3.1.1.2 Pre-processing:

The switch input must be stable before it is given to an IC. Pull-down resistors are connected so that the IC input does not float when the switch is open. When the switch is ON, the input receives logic 1. When the switch is OFF, the input remains logic 0. This pre-processing block is important part as it reads the physical switch state and sends the binary value into the logic circuit.

3.1.1.3 Feature extraction:

Feature extraction in this project means generating equality features from raw binary bits. Each entered bit is compared with the stored bit by XOR gate. The output of XOR is 0 when bits are same and 1 when bits are different. Therefore, NOT gate is used after XOR to generate equality output. Four equality outputs are generated for four password bits.

3.1.1.4 Classification:

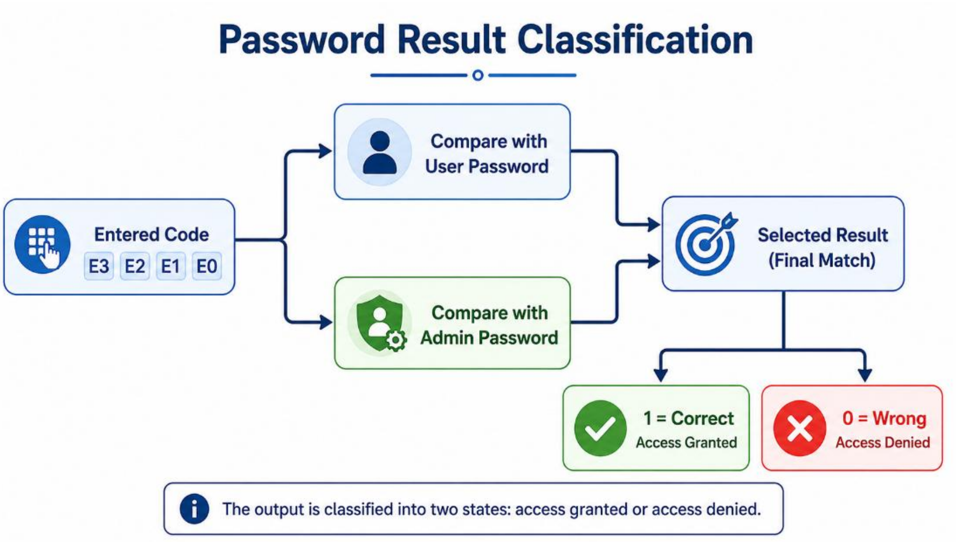

Figure 15 Block Diagram of Classification

3.1.1.4.1 Input Password

The input password is entered with four binary switches. These switches represent E3, E2, E1 and E0. The circuit reads these four bits and compares them with stored password bits. The input can be changed by the user before pressing ENTER.

3.1.1.4.2 Stored Password Path

In the stored password path, we have assigned two password groups. One group is for normal user password and one group is for admin password. These stored passwords are also represented by 4-bit switch combinations. The circuit checks only the selected path according to the mode input.

3.1.1.4.3 Variable

The variable in the circuit is the MODE input. MODE decides whether the normal user match or admin match will be accepted. If MODE is 0, normal user path is selected. If MODE is 1, admin path is selected. This variable makes the circuit dual-mode.

3.1.1.4.4 Output

After all this the resultant signal will be classified into two classes. These classes are whether access granted which is represented by green LED or access denied which is represented by red LED and buzzer. In the output logic, final match equal to 1 indicates correct password and final match equal to 0 indicates wrong password.

3.1.1.4.5 Comparator Logic

Comparator logic is used to detect equality between entered password and preset password. The equation for each equality bit is Xi = NOT(Ei XOR Pi). The complete match equation is Match = X3.X2.X1.X0. The project uses 74LS86 for XOR, 74LS04 for NOT and 74LS08 for cascaded AND logic.

3.1.1.4.6 Sequential latching

Sequential latching is achieved using the 74LS74 D flip-flop. The final match signal is connected at D input of one flip-flop and the ENTER button is connected at clock input. When the user presses ENTER, the output gets latched. RESET clears the flip-flop for a new entry.

3.1.1.4.7 Final Access Control

Final access control is produced after mode selection and flip-flop latching. The final match equation is FM = (User Match . NOT MODE) + (Admin Match . MODE). The inverted final match is used for wrong password path. This output controls the LEDs and buzzer according to password condition.

3.2 Analysis Procedure

In the analysis procedure different components are selected to perform the desired tasks. Here the task is to select those components which are more suitable for the cause and can give better results. The circuit uses 5V TTL logic ICs because they are easy to connect and commonly used in digital logic labs.

3.2.1 Combinational Logic and Sequential Logic

Combinational and sequential logic both have one common issue of noise and stability if input connections are not proper. Combinational logic has a syntax which can be easily understood through Boolean expressions. Sequential logic stores the result and makes the output stable. That is why we had decided to use both combinational logic and D flip-flops.

Table 4 Comparison of Combinational and Sequential Logic

Sr. No	Combinational Logic	Sequential Logic
1.	Output depends only on present inputs.	Output depends on present input and stored state.
2.	Used for password comparison and mode selection.	Used to latch access and alarm output.
3.	Implemented using XOR, NOT, AND and OR gates.	Implemented using 74LS74 D flip-flop.
4.	No memory element is used in this stage.	Memory element is used in this stage.
5.	Faster response to switch changes.	Output changes only at clock pulse.
6.	Can flicker if input changes.	Keeps result stable after ENTER.
7.	Needs correct Boolean expression.	Needs correct clock and reset wiring.
8.	Forms the decision logic.	Forms the final display logic.

3.3 Design of the Project Hardware

In this design procedure we decided that a prototype circuit will be implemented on a large breadboard with attached logic ICs in it. This breadboard model is according to the dimension and spacing of available TTL ICs. The controller is not a microcontroller; instead, the control is performed by ICs 74LS86, 74LS04, 74LS08, 74LS32 and 74LS74. Each IC is powered through 5V and ground rails.

3.4 Design of the Project Software/Algorithm

There are two types of design support used for checking and implementing data from different components of project. We will discuss them one by one how they are used for interfacing of different parts of project.

3.4.1 Proteus simulation and logic testing

A main design support of our device for checking the password is Proteus simulation. The ICs are placed virtually and the input states are changed to check the response of circuit. The simulation helps to verify user match, admin match, final match, access LED and alarm path before wiring on breadboard.

3.4.2 Boolean expression and test procedure

Second design support used in this project is Boolean expression analysis. The expressions are derived from truth table and are implemented through available 2-input gates. The test procedure includes entering correct user password, entering wrong user password, selecting admin mode, entering correct admin password and pressing reset after each output.

3.4.3 Details of Final Working Prototype

The final working prototype of project is in the form of breadboard-based access control system. The prototype is capable of accepting 4-bit password input, selecting user/admin mode, holding output with ENTER.

3.5 Designing of breadboard model:

The prototype breadboard model is made by placing all ICs across the breadboard central gap. The +5V and GND rails are first connected. For each 14-pin IC, pin 14 is connected to +5V and pin 7 is connected to GND. DIP switch outputs are connected to logic inputs through pull-down resistors. LEDs are connected with 330 ohm resistors. The buzzer is connected with the wrong password output path. Every connection is checked step by step as a precaution.

3.6 Summary

This chapter discussed the hardware and logic design of project. The chapter briefly tells us about the simulation implementation and designing of digital architecture used for password verification. The block diagram is working of logic implementation. It also discusses the breadboard modeling of project prototype. This chapter tells us about the detail of wiring and designing of prototype access control circuit. Moreover, it discusses the different tools and their interfacing with different project components in detail.

Chapter 4

TOOLS AND TECHNIQUES

This chapter includes detail about the tools used in the project that will be discussed below. In which two types of design support are described in detail in the software tools

4.1 Hardware Tools

4.1.1 Logic ICs

Logic ICs used in this project are 74LS86, 74LS04, 74LS08, 74LS32 and 74LS74. 74LS86 is a quad XOR gate IC and it is used to compare the entered password bit with stored bit. 74LS04 is a hex inverter and it is used to convert XOR output into equality logic. 74LS08 is a quad 2-input AND gate and it is used to combine equality outputs. 74LS32 is a quad OR gate and it is used to combine selected match paths. 74LS74 is a dual D flip-flop and it is used to latch access and alarm outputs.

4.1.2 Breadboard

The breadboard is used for complete circuit implementation without soldering. It provides rows and columns for connecting IC pins, DIP switches, resistors, LEDs and jumper wires. A large breadboard is preferred because the circuit contains many gates and connections. Proper spacing reduces congestion and makes troubleshooting easier.

4.1.3 LEDs and Buzzer

The LED and buzzer section is used to show the result of the password. Green LED indicates access granted. Red LED indicates access denied. Yellow LED can indicate admin mode. The buzzer rings when wrong password is entered. The LEDs are protected with 330 ohm resistors. Pull-down resistors are used on switches and push buttons.

4.2 Software's and Simulation Tools

4.2.1 Proteus Design Suite

Proteus is used for the simulation purpose. We have to check the password circuit so for that we worked on our circuit by Proteus. It integrates schematic placement, wiring and simulation in an easy-to-use environment where problems and solutions are expressed through logic values. In this project, Proteus is used to verify the digital circuit before practical breadboard implementation.

4.2.2 Truth Table and Boolean Simplification

Truth table is used in our project to stimulate our logic design. This is also for finding the correct password and wrong password using binary values. Secondly, the truth table also helps to derive Boolean expressions. It performs other function like input/output verification, mode selection checking and reduction of logic errors.

4.2.3 Manual Testing

Manual testing is used for the checking purpose on breadboard. We have to find the correct working in our circuit so for that we worked on one section at a time. First comparator outputs were checked, then AND gate match outputs were checked, then mode selection was checked, and finally flip-flop, LEDs and buzzer were tested.

4.3 Summary

This chapter is about the discussion of different software and hardware tools which are being used in this project. The purpose includes the detailed discussion about the software's used for different designing and simulation. These are used for testing the logic for getting the required and accurate results. Proteus is for simulation of circuit. Truth table and Boolean expressions are used for encoding the algorithm of the circuit. Breadboard and ICs are used for final hardware implementation.

Chapter 5

PROJECT RESULTS AND EVALUATION

In this chapter, all the details about the results and the findings are going to be discussed. At the start of the project the things that are being claimed have achieved or not. If not achieved, then what are reasons that the desired results have not achieved.

5.1 Presentation of the findings

A general description of results of the project. Tables/Waveforms/Graphs should be used to describe the results.

5.1.1 Hardware Result:

The breadboard circuit is connected with TTL ICs and 5V supply. The DIP switches are used to give entered password, normal user password and admin password. The mode selector chooses the required password mode. The ENTER button clocks the final result into the flip-flop. The RESET button clears the output. When the entered password is correct, the green LED turns ON. When the entered password is wrong, the red LED turns ON and buzzer rings.

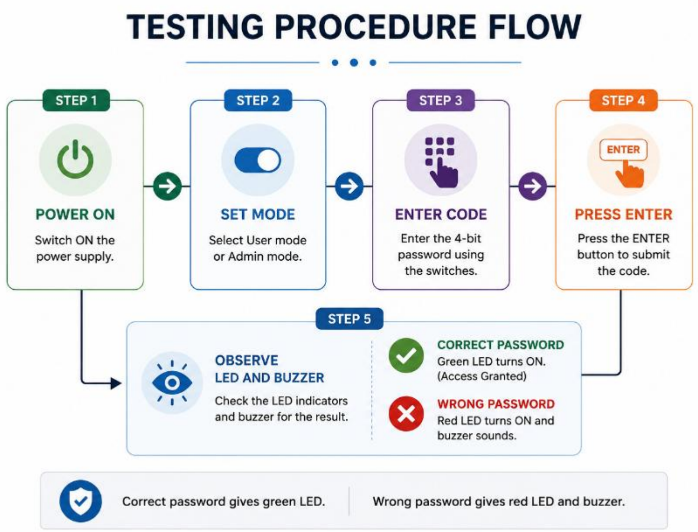

Figure 16 Testing procedure flow

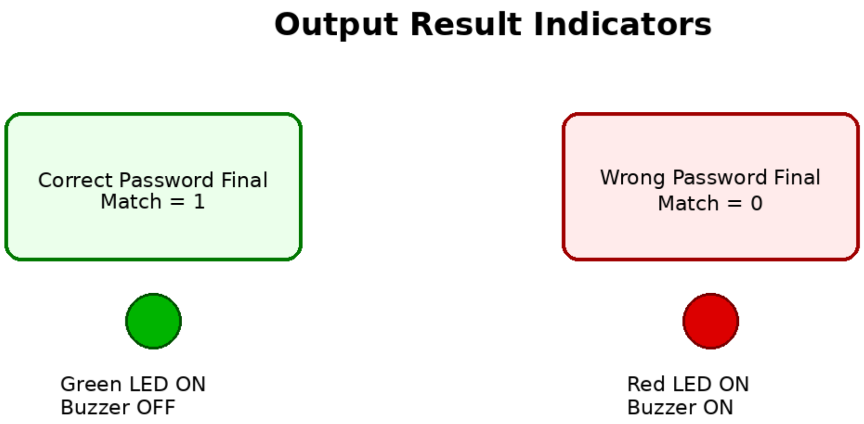

Figure 17 Output result indicators

5.1.2 Simulation Results

5.1.2.1 Test Cases

The test cases for our Password-Based Digital Access Control System are taken from the expected truth table. The test cases contain two classes one is correct password and second one is wrong password. Further they both have user and admin mode cases. The output is verified using green LED, red LED and buzzer condition.

5.1.2.2 Truth Table Outputs

The truth table output shows the result for selected combinations. When all four equality outputs are 1 and the selected mode matches, final match is 1. If any one equality output becomes 0, final match becomes 0 and alarm path is activated after ENTER.

Table 5 Test Cases

Sr no	Input Condition	Expected Output
1	MODE = 0 and entered password = user password	Green LED ON, buzzer OFF
2	MODE = 0 and entered password != user password	Red LED ON, buzzer ON
3	MODE = 1 and entered password = admin password	Green LED ON, admin mode selected
4	MODE = 1 and entered password != admin password	Red LED ON, buzzer ON
5	RESET pressed	All latched outputs cleared
6	ENTER not pressed	Output does not latch

5.1.2.3 Build model

· Select logic ICs

We selected ICs that are used for the logic design and practical implementation. The two main categories are comparator ICs and latching ICs. 74LS86 and 74LS04 are used for bit comparison, 74LS08 and 74LS32 are used for match selection, and 74LS74 is used for output latch.

· Initializing the circuit

Basically, the power supply is first initialized and all ICs are connected with +5V and ground. After this the DIP switches are connected through pull-down resistors. Then each gate output is tested by changing input values.

· Combining operation

The model uses cascaded 2-input AND gates that are used to reduce four equality signals into a single match signal. The mode selection gates choose between user match and admin match. The final OR gate produces final match.

· Add flip-flop block

A flip-flop block is added so that access granted and alarm states remain stable. The ENTER button is connected to the clock and RESET clears the result. This makes the practical circuit more reliable.

5.1.2.4 Testing Parameter

Table 6 Testing Parameter

Sr no	Designing	Working
1	Power supply	5V DC regulated
2	Classes	Two classes: correct and wrong
3	Password length	4 bits
4	Operating modes	Normal user and admin
5	Comparator gate	74LS86 XOR
6	Inverter gate	74LS04 NOT
7	Match gate	74LS08 AND
8	Selection gate	74LS32 OR
9	Latch IC	74LS74 D flip-flop
10	LED resistor	330 ohm
11	Pull-down resistor	10k ohm
12	Correct output	Green LED ON
13	Wrong output	Red LED and buzzer ON
14	Reset	Clears latched outputs
15	Supply convention	Pin 14 = +5V and pin 7 = GND for 14-pin ICs

5.1.2.5 Classification Report

A classification report is a performance measuring metric in a verification problem. In this project, the classification report is generated to check the number of correct access and wrong access decisions. Correct state indicates the percentage of the input combinations that are accepted only when the selected password matches. Wrong state indicates the percentage of the input combinations that are rejected when password does not match. The project gives correct result according to the following logic equations:

User Match = U3eq . U2eq . U1eq . U0eq

Admin Match = A3eq . A2eq . A1eq . A0eq

Final Match = (User Match . NOT MODE) + (Admin Match . MODE)

Alarm = NOT Final Match

5.1.2.6 Confusion matrix

A result matrix is a summary of prediction results on a password verification problem. The number of correct and incorrect outputs are summarized with count values and broken down by each mode. In the user mode, correct user password should generate access granted. In admin mode, correct admin password should generate access granted. In both modes, wrong password should generate access denied and buzzer ON.

Table 7 Result Matrix

Selected Mode	Entered Password	Final Match	Green LED	Red LED/Buzzer
User	Correct user password	1	ON	OFF
User	Wrong user password	0	OFF	ON
Admin	Correct admin password	1	ON	OFF
Admin	Wrong admin password	0	OFF	ON

5.2 Summary

The chapter discusses the results of project. The images above showing the hardware and simulation result of password access control circuit. The gates used in it and how the logic works. The testing history and output states show the efficiency of digital logic implementation. All the results show briefly in this chapter. An evaluation of hardware and simulation results is discussed in this chapter.

Chapter 6

CONCLUSION AND FUTURE WORK

In Pakistan and in many educational laboratories, simple security systems are usually explained by microcontrollers. This Password-Based Digital Access Control System is controlled by digital logic ICs which gives us result about correct and wrong password condition. In PART-1 we covered simulation implementation where we prepared truth table, Boolean expressions and Proteus circuit. In PART-2 we designed the practical prototype with its accurate IC placement and breadboard wiring. We implemented comparator logic, mode selection and flip-flop latching on hardware and tested the result. The input switches were taking the binary password when the user changed them manually. The entered password was cross checked with the selected stored password. The correct password was represented with green LED and the wrong password was represented with red LED and buzzer. This is how the project was working. The project provides a clear demonstration of Digital Logic Design concepts. This access control system makes a relief in the field of learning gate-level security circuits. Furthermore, increasing the password length, adding a keypad, adding debouncing circuit and using seven-segment display can have a more promising result.

References

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[4] Texas Instruments, SN74LS04 Hex Inverter Datasheet.

[5] Texas Instruments, SN74LS08 Quad 2-Input AND Gate Datasheet.

[6] Texas Instruments, SN74LS32 Quad 2-Input OR Gate Datasheet.

[7] Texas Instruments, SN74LS74 Dual D-Type Positive-Edge-Triggered Flip-Flop Datasheet.

[8] Labcenter Electronics, Proteus Design Suite User Documentation.

[9] Digital Logic Design Laboratory Manual, Department of Electrical and Computer Engineering.

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APPENDICES

Appendix - A

Truth Table and Boolean Expressions

Table 8 Single-bit equality truth table

Ei	Pi	XOR Output	Equality Xi
0	0	0	1
0	1	1	0
1	0	1	0
1	1	0	1

Single bit equality: Xi = NOT(Ei XOR Pi)

User match: UM = U3eq . U2eq . U1eq . U0eq

Admin match: AM = A3eq . A2eq . A1eq . A0eq

Final match: FM = (UM . NOT MODE) + (AM . MODE)

Wrong password: W = NOT FM

Appendix – B

Breadboard Wiring Notes

· Connect pin 14 of every 14-pin TTL IC to +5V.
· Connect pin 7 of every 14-pin TTL IC to GND.
· Use 10k ohm pull-down resistor with switch inputs and push buttons.
· Use 330 ohm resistor with every LED.
· Keep wires short and connect one logic section at a time.
· Test XOR/NOT equality outputs before connecting AND gates.
· Test user match and admin match separately before connecting mode selection.
· Connect ENTER to the clock input of 74LS74 and RESET to reset input.
· Keep buzzer connected only to wrong password output path.

Appendix – C

Final Project Implementation

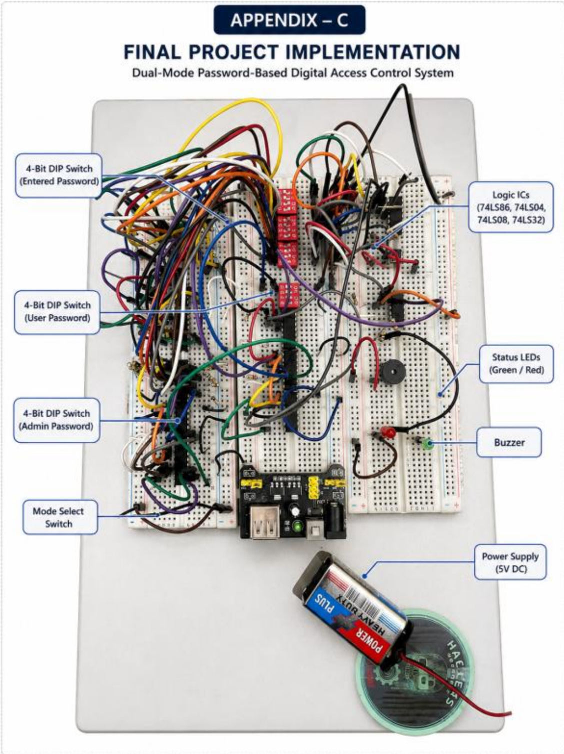

Figure 18 Final Project Implementation

Appendix – D

Final Project Implementation (proteus)

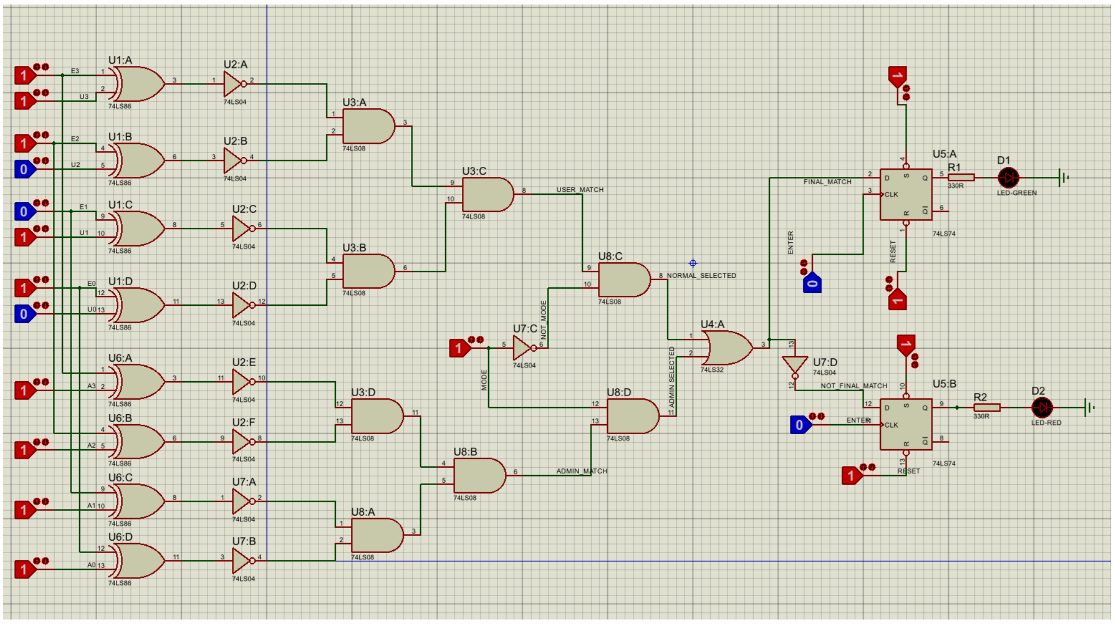

Figure 20 Final Project Implementation (proteus)